Two Locations, Two Rates, One Spreadsheet Error

Prepared for PABS. Illustrative example with fictional figures, built to be checked by hand. This example is a worked illustration, not an engine run.

The problem

Franchise royalties look simple until locations stop sharing one rate. A first-year location on a stepped royalty, a marketing fund computed on a different base, a rate that steps up on an anniversary: each one is a place where a spreadsheet quietly applies the standard term to the wrong location.

Location (fictional)	Monthly gross sales	Contract terms
Riverside	$86,400	Royalty 6 percent; marketing fund 2 percent
Hillcrest, first year	$52,300	Stepped royalty 5 percent; marketing fund 2 percent

Computed correctly, Hillcrest owes a royalty of $2,615.00 for the month. A spreadsheet that applies the standard 6 percent rate to every location bills $3,138.00 instead. The franchisee is overbilled $523.00 in one month, at one location, and the same habit repeats at every stepped-rate location in the network until someone disputes a statement.

> Nobody mistyped a number. The spreadsheet did exactly what it was built to do: one rate, applied everywhere. The contract said otherwise, per location, with dates attached.

The same example in RevRule

In RevRule, each location's royalty and marketing fund terms are stored as rules with their rates and effective periods. The stepped first-year rate applies automatically during year one and steps up on schedule; nobody retypes a rate at period end, so the standard rate cannot leak onto the wrong location.

Amounts owed for the month	Royalty	Marketing fund
Riverside	$5,184.00	$1,728.00
Hillcrest (5 percent year-one rate)	$2,615.00	$1,046.00
Network total	$7,799.00	$2,774.00

Every amount is recorded in a hash-chained ledger alongside the rule version that produced it, queryable by location, participant, or entry type and returnable in full as JSON. The $523.00 error above cannot occur, because the rate is never retyped. For a firm that white-labels royalty calculation to CPA firms, that record is the deliverable: the franchisor, the franchisee, and the accountant are all looking at the same computation. (Illustrative calculation on the fictional figures from page 1.)

RevRule does not hold funds, execute payments, or collect royalties. It computes what is owed under the contract for money already collected.

Full product details: https://payloadhq.github.io/revrule.html